

When Not to Learn: Evidence-Gated Residuals for LR-FHSS D2S IoT

A real-data negative result on BLACK KITE TLE histories

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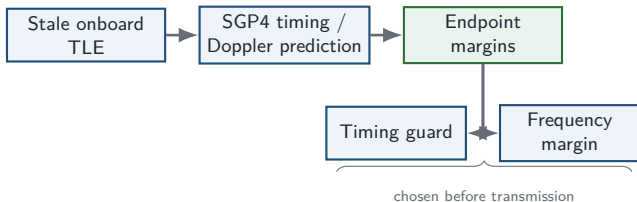
Affiliation omitted for double-blind review

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Advisor review / workshop draft

**Residual ML must earn deployment; real evidence retains
SGP4.**

Motivation and System Problem



The endpoint has orbital metadata, not a link outcome. It must size timing and frequency margins before the burst.

Context: LR-FHSS direct-to-satellite operation couples orbital prediction error to endpoint resource choices.

Endpoint question

Should a resource-constrained terminal deploy a learned inter-TLE residual correction over its available SGP4 baseline?

The control decision is pre-transmission: stale TLE to prediction to endpoint margins. Selected context: Semtech

AN1200.64; Ullah et al., OJ-COMS 2024.

Research Gap

LR-FHSS link / capacity

Reliability, coverage, collision, and direct-to-satellite capacity studies.

Semtech AN1200.64; Ullah WCL; LoRa D2S uplink policies

NTN synchronization

Receiver-side synchronization and NTN mobility robustness.

3GPP TR 38.821; Ullah OJ-COMS

Orbit / physics-ML correction

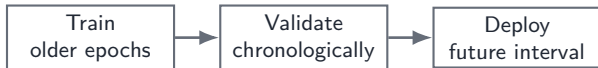
SGP4, orbit prediction, differentiable propagation, and residual modeling.

Hoots-Roehrich; Peng et al.; differentiable SGP4; ML orbit survey

Gap. Prior work predicts, receives, or schedules; it does not decide whether a resource-constrained endpoint should deploy learned inter-TLE correction over its available SGP4 baseline.

The missing object is a deployment rule, not another predictor leaderboard.

Hypothesis and Failure Risk



domain shift risk



training fit can look plausible



deployment can be worse

Hypothesis

Residuals may be learnable.

Failure risk

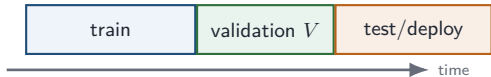
Chronological shift can make always-on ML worse than SGP4.

A plausible learner remains unsafe until it wins on held-out chronological evidence.

Learning enters as an untrusted candidate.

Method: Evidence Gate

$$G = \mathbf{1}[\text{MAE}_{\text{ml}}(V) < \gamma \text{MAE}_{\text{phys}}(V)]$$



chronological split; no future TLE leakage

Training fits the candidates. G is fixed on validation; the test segment only reports consequences. MAE is the gate loss; RMSE is a diagnostic, not part of the rule.

What γ does

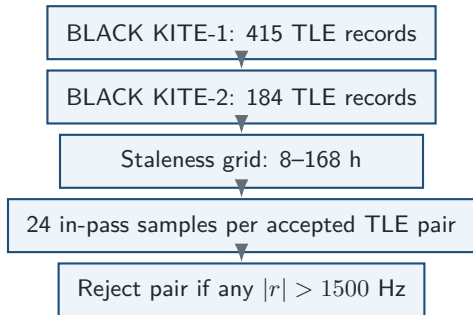
Smaller γ demands a clearer held-out gain. It reduces false-open risk at the cost of missed-open opportunities.

$\gamma = 0.95$ in all experiments.

Audit rule, not a new predictor. The deployable baseline remains SGP4 unless the learner earns the branch.

The rule decides whether learned correction may deploy over the available baseline.

Experimental Protocol



The real claim is conditional on accepted non-outlier pairs;
 $|r| > 1500$ Hz pairs are excluded and left for a maneuver/outlier extension.

Chronological split



Reference boundary

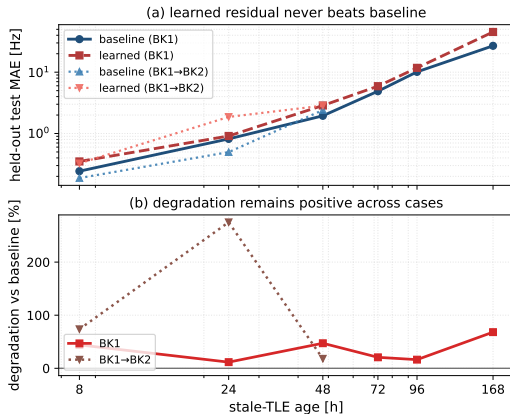
`reference_is_measured_truth=false`. All real results are model-derived inter-TLE residuals.

Candidate models

Ridge, random forest, gradient boosting, and small MLP selected on validation.

The real-data protocol tests deployable chronological utility, not measured Doppler truth.

Main Real Result



All reported BLACK KITE point estimates close the gate; deploy SGP4 / never-learn.

Reported point estimates are worse in every row; BK1 degradation 11.6%–68.1%. Cross-satellite rows (up to 275.1%) are a limited transfer check; per-split counts were not preserved. Point estimates under the leakage-free protocol; not pair-clustered inference.

Why a Closed Gate Matters

Always learn

Deploys the learned branch by default.

On BLACK KITE, this increases residual error.

Never learn

Retains the SGP4 stale-TLE baseline.

On BLACK KITE, this is the selected safe output.

Evidence Gate

Tests the learned branch on chronological validation.

On BLACK KITE, it chooses the same output as never-learn.

At BK1 168 h, $26.9 \rightarrow 45.3$ Hz is still $\ll F_{\text{tol}} = 500$ Hz: this is a refusal/audit result, not an energy-win claim.

Gate value here is auditability: it blocks unsupported always-on ML, not guaranteed improvement.

Controlled Synthetic Mechanism Check



Controlled software-only check.

Synthetic cases open only when a dominant systematic residual is present.

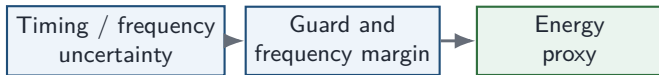
Noise dominated: closed.

Marginal gain below γ : closed.

Dominant systematic residual: open.

Synthetic evidence checks mechanism only.

Endpoint Implications



coverage equations map residual uncertainty
to software resource proxies

Software-only control proxy; not a packet result.
Proxy scale is illustrative: it is not the BLACK KITE residual
scale and not a link result.

Design implication

A learner that does not win the gate should not consume extra guard, frequency margin, or energy budget.

Audit implication

The deployed policy must record why the learned branch was allowed or blocked.

The proxy connects residual evidence to endpoint budget pressure without field-result evidence.

Limitations and Next Campaign

Current limitations

- ▶ Two BLACK KITE satellites
- ▶ Model-derived reference only; no measured Doppler truth
- ▶ Oscillator and channel effects may dominate practical endpoints
- ▶ MAE gate is not tail-aware

Next software campaign

- ▶ Multi-satellite TLE stress matrix
- ▶ Residual autocorrelation and learnability analysis
- ▶ Stronger baselines
- ▶ Tail-aware or cost-aware gate

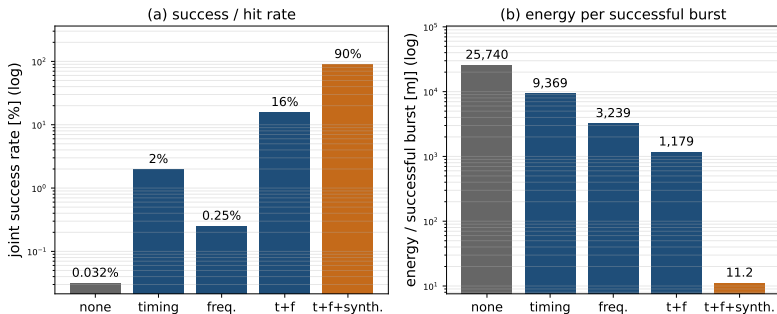
Broader deployment value requires broader chronological evidence and a loss beyond MAE.

Contributions and Takeaway

1. **Real-data negative finding.** On BLACK KITE TLE history, the reported inter-TLE point estimates are worse than SGP4 in every row.
2. **Evidence-gated deploy/refuse audit rule.** G is decided on chronological validation; on real BLACK KITE data the gate closes and deployment equals SGP4 / never-learn.
3. **Secondary: illustrative endpoint-budget proxy.** Timing, frequency, and energy figures are software-only coverage proxies, not packet or field results.

**Do not trust residual ML by default.
Let chronological evidence decide deployment.**

Backup: Endpoint Proxy Ablation



Detailed software-only proxy ablation. The orange right-most bar (t+f+synth.) is the synthetic gate-open branch, not a BLACK KITE improvement.

Backup: Counts and Gate Sensitivity

Experimental counts

BLACK KITE-1	415 TLE records
BLACK KITE-2	184 TLE records
Samples	24 per accepted pair
Split	chronological 60/20/20
Staleness	8–168 h
Reject rule	any $ r > 1500$ Hz

Gate interpretation

- ▶ Lower γ demands larger validation gain.
- ▶ False-open risk: deploying a harmful learned branch.
- ▶ Missed-open risk: retaining SGP4 despite real systematic gain.
- ▶ Real BLACK KITE rows close under the tested settings.

The sensitivity question is policy conservatism, not predictor novelty.

Backup: Software Extension Diagnostics

What the extension confirms

- ▶ Lightweight bias and baseline variants still do not beat SGP4 on BLACK KITE.
- ▶ The real MAE gate remains closed in every reported row.

Median bias, ridge, random forest, gradient boosting, and small MLP; summary-level audit of committed artifacts, not fresh retraining.

What it does not establish

- ▶ A tail-aware real gate needs per-sample prediction export; not claimed here.
- ▶ The multi-satellite pipeline is prepared as future work; no new generalization claim.

All rows are software-only, model-derived inter-TLE results.

The extension reinforces the same safe decision; it does not widen the claim.

Backup: What This Work Does Not Claim

- ▶ No measured Doppler truth; the reference is a later TLE propagated by SGP4.
- ▶ No packet-level, decoding, error-rate, or receiver-acknowledgement outcome.
- ▶ No over-the-air transmission and no on-orbit result.
- ▶ The synthetic gate-open case is a mechanism check, not a real-data improvement.
- ▶ Endpoint proxy curves are software-only; they are not a link-layer result.

The claim is a deployment decision on model-derived inter-TLE residuals: on BLACK KITE, keep SGP4 / never-learn.